

Project Title

Smart Lender – Loan Eligibility Prediction System

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INTRODUCTION

Smart Lender is an intelligent Machine Learning-based web application developed to predict the eligibility of loan applicants. The system helps banks and financial institutions make faster, more accurate, and data-driven lending decisions by analyzing applicant information and predicting whether a loan should be approved or rejected.

The application leverages historical loan data and multiple Machine Learning classification algorithms to identify the most suitable model for loan approval prediction. The best-performing model is integrated into a Flask-based web application that enables users to enter applicant details and receive instant loan eligibility results.

The system evaluates various factors such as applicant income, co-applicant income, loan amount, loan term, credit history, education, employment status, marital status, number of dependents, property area, and other relevant financial details before generating a prediction.

To enhance the user experience, the application provides a modern and responsive interface where users can submit loan applications, view approval or rejection results on a separate page, and receive helpful suggestions when a loan application is rejected. This feature makes the system more informative and user-friendly by guiding applicants on improving their chances of future loan approval.